(1)

(a) In order to maximize selectivity, a low conversion of ethane should be targeted, since too much in the reactor would propel the second reaction. You will need a recycle stream and separator, since there will be a lot of unreacted ethane.

(b) $\xrightarrow{n_1 \text{ (ethane)}}$ $\xrightarrow[n_2 (Cl_2)]{}$ Rx $\longrightarrow n_3$ (ethane)
$\longrightarrow n_4$ $(Cl_2)$
$\longrightarrow n_5$ $(C_2H_5Cl)$
$\longrightarrow n_6$ $(C_2H_4Cl_2)$
$\longrightarrow n_7$ $(HCl)$

Unks: 7 ($n_1 \to n_7$)
Atomic Balances: 3 (C, H, Cl)
$7 - 3 = 4$

(c) Basis: 100 moles, moles ethane reacted $= 100 \times .15 = 15 = n_{mono} + n_{Di}$,
$\frac{n_{mono}}{n_{Di}} = 14$, $n_{mono} = 14$, $14 n_{mono} + n_{Di} = 15$, $n_{Di} = 1$ mol $C_2H_4Cl_2$,
14 moles of $C_2H_5Cl$, total $Cl_2$ moles $= (1 \times n_{mono}) + (2 \times n_{Di}) = 16$ mol
Feed ratio $= \frac{n_{Cl_2}}{n_{C_2H_6}} = \frac{16}{100} = .16$, Fractional Yield $= \frac{14}{16} = 87.5\%$

(d) It likely means there was further chlorination, getting $C_2H_3Cl_3$ in the product.

(2) Assume 1 mol G fed: $x$ = G reacted, $n_G = 1-x$, $n_w = 5-3x$, $n_c = 3x$, $n_{H_2} = 7x$
total moles $= 6+6x$, $\frac{7x}{6+6x} = .54$, $x = .86$, $X_g = 86.7\%$.
$Q_p = \frac{(.23 \times 1.2)(.54 \times 1.2)}{(.012 \times 1.2)(.216 \times 1.2)} = .165$, the reactor is not at equilibrium.

Either increase temperature (since it should be endothermic) or increase residence time. $\frac{(3x)^3(7x)^7}{(1-x)(5-3x)} \times \frac{P^6}{(6+6x)^6} = 55$, Solver says $\boxed{.9}$ equilibrium conversion

3. p-xylene + $O_2 \to$ TA + $H_2O$, high selectivity, but loses 10% acetic acid to oxidation.
Cannie's process: $C_8H_{10} + 3H_2O_2 \to C_8H_6O_4 + 4H_2O$, atom economy $= \frac{166}{106+122} = 74.8\%$
Cannie's yields and selectivity (70%, 90% respectively) are lower than the conventional process. Also, $H_2O_2$ is more expensive.

4. ($R_1$) $CH_3OH + NH_3 \rightarrow CH_3NH_2 + H_2O$

($R_2$) $CH_3NH_2 + CH_3OH \rightarrow (CH_3)_2NH + H_2O$

Total feed = 200 mol

$200 \times .2 = 40$ mol $NH_3$, $200 \times .094 = 18.8$ mol MOH, $200 \times .186 =$ 37.2 mol MA, $200 \times .104 = 21.8$ mol DMA, $H_2O = 82.2$ mol

a. $\frac{100 - 18.8}{100} = .812$

b. $\frac{37.2}{100} = .372$

c. $37.2 / 21.8 = 1.71$

d. $100 - 40 = 60$ gmol/hr

e. 21.8 gmol/hr

$$K_{y1} = \frac{(.186)(.411)}{(.094)(.2)} = 4.07$$

$$K_{y2} = \frac{(.109)(.411)}{(.186)(.094)} = 2.5$$

The reactions are basically at equilibrium